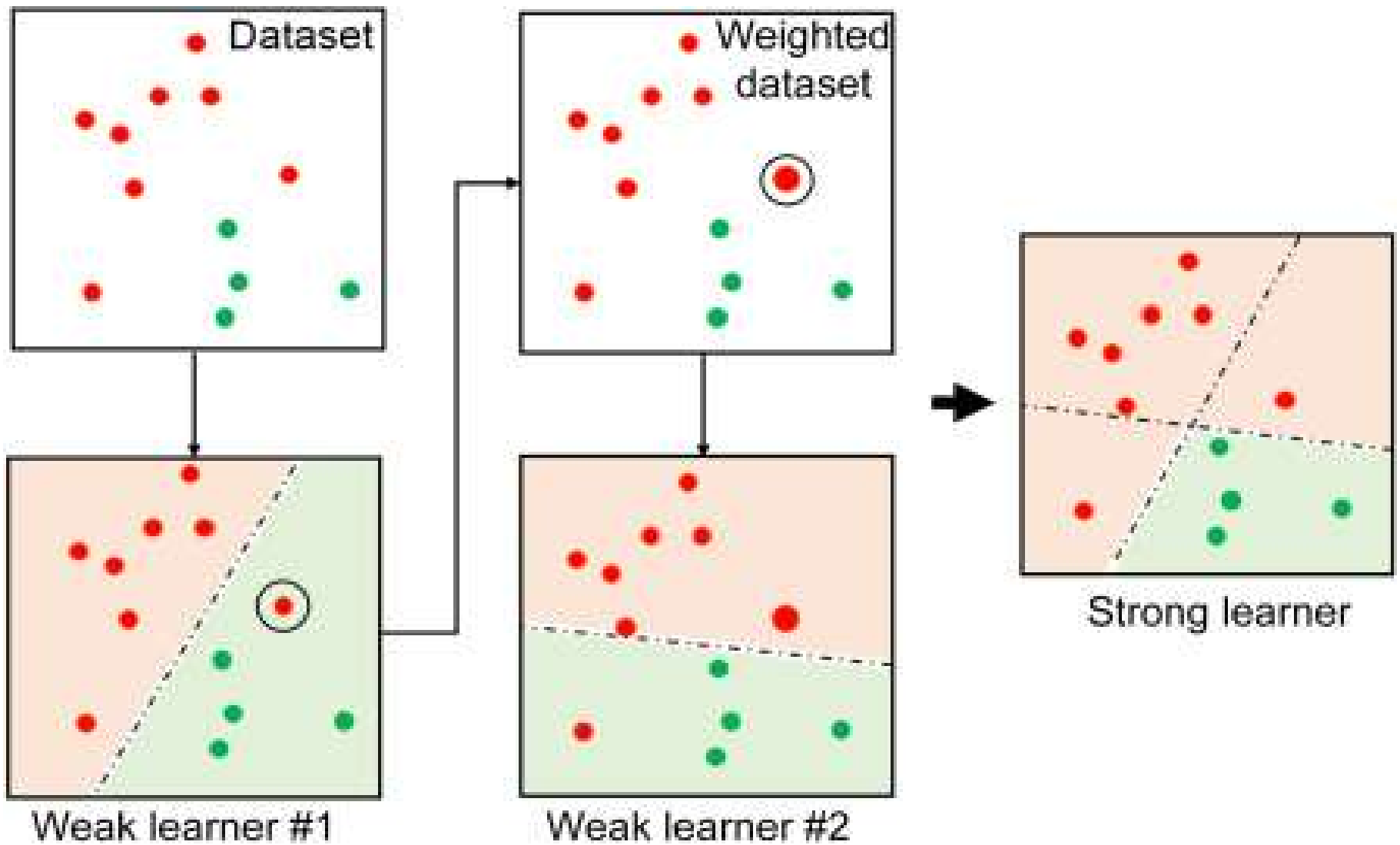


What is AdaBoost?

- AdaBoost stands for Adaptive Boosting.
- It is a machine learning boosting algorithm.
- Combines many weak models to create one strong model.
- Mostly used for classification and regression tasks.

How AdaBoost Works

- Step 1: Train the first weak model.
- Step 2: Find incorrect predictions.
- Step 3: Give more weight to difficult data points.
- Step 4: Train the next model focusing on mistakes.
- Step 5: Combine all models for final prediction.



Advantages and Disadvantages

- Advantages:
 - - High prediction accuracy
 - - Reduces bias and improves learning
- Disadvantages:
 - - Sensitive to noisy data
 - - Can be slower with many models

Simple Real-Life Example

- Imagine a teacher helping students.
- The teacher focuses more on weak students after each test.
- Over time, all students improve.
- AdaBoost works similarly by focusing on wrong predictions.